

Chapter 4

Connecting, Exploring, Uncovering, and Building Partnerships



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4.1 Introduction

The landscape of University-Industry (UI) relations is rich and multifaceted, encompassing pivotal components like research collaborations, innovation, technology translation, experiential learning, and even executive education. Indeed, in its 2011 White Paper, the Network of Academic Corporate Relations Officers (NACRO) identified five “Essential Elements of a Successful Twenty-First Century University Corporate Relations Program,” including research development, campus coordination, and aligned interests. Yet forging long-term impactful relationships between academia and industry proves to be more challenging because the two sectors, industry and academia, often differ dramatically in their missions (for profit vs. nonprofit), drivers, constraints, stakeholders, and decision-making organizational structures. For example, companies typically operate under top-down directives, whereas universities thrive on a bottom-up approach, granting faculty considerable autonomy over research and curricular decisions. Moreover, it is often said that “you know one company or university, you know only one company or university.” The stark contrast in organizational dynamics and the individuality of each organization present both opportunities and hurdles in fostering meaningful connections between the two sectors. Indeed, navigating the complexities inherent in UI partnerships demands a deliberate and strategic approach on the part of each partner. It requires a nuanced understanding of the divergent cultures and motivations at play.

In essence, connecting, learning, and cultivating relationships between universities and companies is a multifaceted endeavor that demands careful attention and intentional effort. Nonetheless, the effort is worthwhile, as the benefits for academia

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and industry take many forms, including access to funding that supports impactful research collaborations, the transfer of innovative technologies to the market, creating a culture of entrepreneurship, training executives versed in emerging technologies with societal impacts, providing relevant experiential learning opportunities for students and the development of a skilled talent pool.

In this chapter, we provide a few examples of practices to connect, explore, and uncover organizational strengths and alignments, and thus build relationships. We share examples drawn from connecting, uncovering alignments, and building upon them to drive meaningful impact. The examples will highlight the inherent nonlinearity of building UI partnerships, extending beyond the framework of our three recommended practices.

4.2 Opportunities and Challenges

Starting a new UI partnership may seem complex, but breaking the process into clear, actionable steps unlocks opportunities for success and fulfillment: identifying the right stakeholders to initiate conversations about partnering, defining the partnership's foundation, and structuring the relationship to create mutual value.

4.2.1 *Connecting*

It seems obvious that, as in any relationship, connecting with the right people within the organization is crucial. For the university, the best external the best industry stakeholders will be those already primed to value the benefits of UI relations to advancing innovation. They may also be willing to navigate organizational complexities. These stakeholders may or may not be ultimately directly involved in all aspects of the relationship but they will be meaningfully impacted by the success of the relationship. They are therefore ready to leverage their influence towards growing the relationship. Champions of UI relations are readily identifiable because they show their interests by participating in academic symposia and industry advisory boards, attending UI professional events (such as UIDP, NACRO, the University Industry Innovation Network, and the Government-University-Industry Philanthropy Research Roundtable), serving as professors of practice, participating in foreign trade missions that showcase university research and capabilities, and engaging alumni.

Take the case of engaging alumni to initiate partnerships. Alumni networks are an effective and strategic approach to cultivate industry partnerships, particularly as alumni are frequently well-acquainted with the university's research strengths. Moreover, alumni with deep ties to both the academic institution and industry

provide valuable insights, broad networks and enhanced credibility, positioning them as facilitators and ambassadors of collaboration between the two sectors. By strategically engaging alumni, universities can unlock key opportunities to initiate and strengthen industry partnerships in research. Alumni can act as advocates, connectors, sponsors, and collaborators that can foster sustained industry relationships. The following list provides examples of ways to engage industry representatives, in general, and alumni, in particular, to help with university-industry partnerships:

Advisory Boards and Roundtables Form industry-specific advisory boards and roundtable discussions that bring together alumni, faculty researchers, and corporate leaders to explore research synergies and industry challenges. These forums foster partnerships by allowing alumni to share their expertise, insights, and networks.

Research Collaboration and Funding Encourage alumni to propose research projects and funding opportunities within their companies. Alumni can play a pivotal role as advocates for university research collaborations, bridging academia and industry.

Mentorship Programs Develop structured alumni mentorship programs for students and early career researchers, providing alumni the opportunity to engage with the next generation of innovators. Track success stories from these mentorships to attract more partners.

Company-Based Alumni Cohorts Help organize company alumni-employees around their affinity for their alma mater and the interests of their company. Collectively, the alumni-employee group can both look out for the interests of the company and advocate for the value proposition of the university, to help grow a strategic partnership with benefits to all stakeholders.

Engage Alumni Entrepreneurs Identify alumni entrepreneurs and connect them with university researchers working on complementary projects to foster innovation and collaboration. These collaborations can lead to joint ventures, start-ups, or sponsored research, aligning academic expertise with market needs.

Start-Up Showcases and Hackathons Create opportunities for alumni to participate in start-up showcases, hackathons or innovation challenges, so they can engage directly with graduate students, postdocs, and faculty researchers while exploring potential collaboration or investment opportunities.

Executive Education and Development Programs Develop industry-specific executive education programs that leverage alumni expertise, offering them a platform to contribute while strengthening ties between the university and industry sectors.

A Case for Participating in Government Trade Missions

Princeton has found value in participating with the Governor of New Jersey in State trade missions organized through the ChooseNJ organization, and has participated in trips to India, Israel, Ireland, the UK, and South Asia (Japan, South Korea and Taiwan) since 2018. Though trade missions require considerable investment of time, the University has reaped tangible benefits, forging meaningful relationships through these engagements.

1. Brings awareness of Princeton's expertise to the State government.
2. Brings awareness of Princeton expertise to the countries visited.
3. Connections to industry and innovation groups in the country visited, organized by the trade mission visits.
4. Partnership prospects with universities in the country, which in some cases can be very attractive to companies in those countries, to think about three-way collaborations.
5. Connections with the other State groups traveling on the trade mission (businesses, government and university representatives, etc.) who may have similar interests.

4.3 Options, Solutions, and Impact

In this section, we make the case for (1) leveraging technology platforms, databases and tools together with corporate engagement professional insight to uncover possibilities for alignment in order to (2) build strategic relationships.

4.3.1 Exploring-Uncovering

Exploring and uncovering the aligned interests of a company and university in order to build a UI collaboration is an intricate assignment. Part of the complexity includes specificity vs. breadth of knowledge; companies are often seeking fairly narrow areas of expertise to solve specific problems rather than seeking out broader subject matter expertise, whereas faculty build their research portfolios on broad subject matter expertise and tackling specific problems as part of their research programs. Moreover, even two companies in the same general industry will likely have different priorities around the same subjects. In the authors' experience there is no single solution or tool that provides perfect technological alignment. Instead the most efficient and effective approach to identify intersecting interests of the university and the company is the use of a variety of technology platforms, databases, and websites in combination with the human insights (that of university corporate engagement professionals (UCEP), technology licensing officers, as well as deans and department chairs).

Databases such as Pivot Interactives, Elsevier's Pure, Academic Analytics' Discovery, and companies such as FirstIgnite, PreScouter, Pillar Science and HaloScience are among a growing number of resources available to do a first pass at matching the expertise on a university campus with the interests of a company. Each of these resources specializes in different aspects of the matchmaking process. For example, Pure can pull up faculty profiles based on keyword searches, whereas HaloScience offers faculty pre-vetted industry projects to submit proposals to. While these technology tools are powerful and can be helpful, they depend on the end-user's access, expertise and familiarity with the tools; novice use may either cast too wide or too narrow a net, potentially missing out on specific researchers whose work would be relevant to a particular company's interest. Leveraging technological platforms can streamline the initial search and identification process. These tools can analyze vast amounts of data, identify potential matches based on research interests and past projects, and provide a comprehensive view of possible collaborators. However, it is crucial to customize these tools to ensure they are tailored to the specific needs of the institution and industry partners. By refining search parameters and using advanced analytics, these platforms can provide more precise and relevant matches.¹

Given the constraints of technology tools, the role of University Corporate Engagement Professionals (UCEP) is indispensable in the UI collaboration process. These professionals bring a deep understanding of the academic landscape, the nuances of departmental research, and the strategic goals of the institution. Their expertise allows them to interpret data from technological platforms effectively, contextualize potential matches, and facilitate meaningful introductions. By combining their knowledge with the insights provided by technological tools, engagement professionals can identify and cultivate relationships that are truly aligned, avoid exhibiting biases influenced by the scope of their portfolio or their academic school or department affiliation, and miss opportunities due to misaligned interests. A significant role of the engagement professionals is staying updated on the latest developments in both academia and industry to understand emerging trends and evolving needs. Their regular training and professional development can help them refine their skills and reduce biases, ensuring they are equipped to manage diverse portfolios and collaborate across various disciplines. Thus, the solutions to creating effective UI relationships require a balanced combination of technological platform tools and the expertise of institutional engagement professionals to ensure good matches and identify topical alignments as the foundation of the relationship.

Additionally, creating opportunities for organic connections between industry and academic researchers is also an effective approach to exploring aligned interests. For example, structured but flexible frameworks for engagement can enhance the effectiveness of UI collaborations. This could involve establishing advisory boards with representatives from both academia and industry, hosting regular

¹ Note that the authors are experienced but not expert users of the resources mentioned, and encourage interested parties to contact the vendors of the resources to learn more about the full range of capabilities of each tool.

networking events and symposiums, and encouraging cross-sector collaborations. These initiatives can create a dynamic ecosystem where stakeholders continuously interact, share insights, and identify new opportunities for collaboration.

In short, while there is no universal solution to identifying aligned interests in UI collaborations, a thoughtful combination of advanced technological tools and skilled institutional engagement professionals can significantly enhance the process. By refining their strategies and fostering an adaptable and interactive environment, universities and industries can forge resilient partnerships that spur innovation and mutual growth.

4.3.2 Building a Relationship Continuum and Strategic Partnership

To summarize the main takeaways in the March 2021 issue of the University Industry Innovation Magazine (UIIM) focused on Strategic Partnerships with Industry, any UI relationship that aims to become a strategic partnership will be built on trust between partners, mutually beneficial goals, consistent points of contact, strategic roadmapping, long-term planning, setting and accounting of annual objectives, and adjusting, pivoting, and growing. A strategic relationship does not necessarily have to involve the whole institution; in fact, it might be easier to start with a specific focus that is later leveraged and grows over time into a larger engagement, as there is no one-size-fits-all approach.

At a minimum, a strategic partnership requires four things: (1) points of contact (POCs) at both organizations, (2) an agreed-to vision and objectives, (3) a roadmap with regular review of the partnership, and (4) accountability on the part of each partner. The industry and academic POCs come to an understanding of each organization's needs in order to articulate and construct the vision for the engagement. The POCs create a roadmap that includes pinpointing the short-, medium-, and long-term objectives of each organization as part of the partnership, with the understanding that these objectives might be different for the company and the university. Regular check-ins and review of the activities enables each partner to be accountable to specific, measurable, achievable, relevant, and time-bound (SMART) goals. An initial goal might be that “over the course of the next year, the partners will identify, execute and embark on at least one translational research collaboration; at the end of the year, the POCs will review what was achieved and assess the satisfaction of each partner, adjust, improve, grow or pivot for the next year.”

Knowing what makes a partnership strategic is important—but the real challenge lies in building one. There is no single way to develop a relationship into a strategic one. In most cases, the plan starts out small as a project between a faculty member and industry researcher that tests the compatibility of the university and company—the organizations' researchers as well as the contractual teams. After the success of the initial project, and with focused attention and time, the university and company

begin to explore shared interests beyond the initial project, laying the groundwork to develop a deeper strategic relationship. At times, the faculty member leading the project is well-positioned to guide these discussions on behalf of the university. However, when a relationship begins to expand beyond the faculty member's primary interests, support from a UCEP is essential. The UCEP can lead broader engagement discussions with company representatives, allowing the faculty member to concentrate on research and technology development. Roles within the partnership become well-defined. The UCEP becomes the university's POC for the relationship, representing the interests of the initial faculty as well as new engagements at the university; the UCEP will seek to grow and steer the conversations with their company counterpart POC into a strategic relationship framework. While the specifics of each strategic partnership will vary, they will all include key points of contact, a shared vision and mission, and a commitment to a mutually agreed-upon strategy.

4.4 Case Studies

This section illustrates two approaches for creating partnerships, the first evolutionary and a second derived from trade missions.

4.4.1 *Evolution of a Strategic Partnership*

Industry partnerships that grow to become strategic usually start incrementally through one-off individual collaborations. Even with the strong CEO support, a strategic relationship is not assured. Success requires intentional management by company and university points of contact to establish a shared vision and objectives, develop a clear roadmap, conduct regular reviews, and ensure mutual accountability. As an example, several years ago a CEO, leading a life sciences company, approached the university with the aim of forming a strategic relationship. His team provided a detailed problem statement to spark collaboration with university researchers. However, the focus was on applied research rather than on fundamental research. Given the company's focus on product development, the initial proposal did not align with the interest of faculty researchers. But rather than ending the engagement, we sowed the seeds for longer-term horizon thinking and provided an opportunity to pivot to better aligned interests.

Princeton emphasized the value proposition of its fundamental research and how it could address complex challenges, potentially leading to breakthroughs. This resonated with the CEO, who sought to foster a culture of innovation within his organization. Collaborating with the company's head of research, Princeton established a structured process to identify challenges that aligned with both the company's key performance indicators and Princeton's research strengths. A Memorandum

of Understanding and a Master Agreement were negotiated which would streamline future collaborations. The shared objective was to establish a system that connected scientists from both organizations, enhancing mutual understanding of research capabilities while efficiently identifying actionable research opportunities. Initially, virtual workshops on broad topics of interest to the company, followed by in-person faculty and company researcher meetings, laid the groundwork for initiating collaborations. The process has evolved, strengthening ties between company segment leads and multiple faculty labs, leading to more than half a dozen collaborations. Campus and lab visits reinforced these collaborations, further building ties between faculty, graduate students, and industry professionals. The engagement has also facilitated student involvement in industry-facing events.

4.4.2 UI Relationships with Japanese Corporations

We offer two examples in this case study. The first partnership began as a long-standing but unstructured relationship with a major automotive corporation that was funding a series of individual research projects. Funding came from multiple branches of the corporation, presenting a range of opportunities, though initially with limited coordination among its many subsidiaries. A master agreement, set up with one company in one specific vertical, was unknown and unusable by other companies in the corporation. Then, in early 2023, the parent corporation underwent internal reflection on its long-term approach to horizon research and its reliance on academic relationships. Princeton University was the first university in the USA that the Corporation engaged with. Coincidentally, the State of New Jersey was planning a trade mission to East Asia around the time when UIDP was hosting a conference in Tokyo. Princeton's participation in both provided an opportunity to include the corporation in both events in October 2023, and propelled the relationship to a new level. The partnership with the parent corporation had been multi-pronged but did not yet have all the aforementioned hallmarks of a strategic relationship; the opportunity to meet with them in Japan as part of the trade mission, solidified the relationship by creating points of contacts, agreed-to vision and objectives, a roadmap with regular review of the partnership, and accountability on the part of each partner.

Our second example is a new relationship with a major heavy industry Japanese corporation that began with meetings facilitated by the State of New Jersey trade mission to East Asia in 2023. Preparations for the meeting in Japan included preliminary calls to build a productive agenda and find thematic alignment; Princeton and corporation POCs relied heavily on the "Research with Princeton" database. Very quickly, the corporation embarked on sponsored projects with faculty, and the POCs are actively engaged to create a small strategic partnership that includes the aforementioned hallmarks.

4.5 Takeaways and Future Strategies

The objective of this chapter was to provide a framework for thinking about connecting, exploring, uncovering, and building UI relationships that are strategic. The framework outlines the key challenges, underscores the importance of alignment and dedicated relationship stewards, identifies concrete opportunities to leverage, and highlights the defining characteristics of strategic partnerships.

1. **Challenges in Collaboration:** Long-term partnerships between universities and industry (UI) are inherently complex, shaped by fundamental differences in missions, structures, and decision-making.
2. **Aligning Interest:** Identifying aligned interests between academia and industry is often challenging, but using data-driven tools and platforms as well as University Corporate Engagement Professionals (UCEP) can significantly enhance this process. Data-driven tools (e.g., Clarivate Pivot, Elsevier Pure, Pillar Science) help identify shared interests and expertise, facilitating more efficient matchmaking.
3. **Points of Contact (POCs):** Designating clear points of contact (POCs) from both university and industry partners is crucial for managing the relationship effectively. The POCs facilitate communication and align interests and manage the relationship effectively.
4. **Unlocking Collaboration Opportunities—Championing Relationships Through Alumni and Trade Missions:** Alumni can play a pivotal role in bridging the gap between academia and industry that might otherwise be difficult to establish. Engaging alumni, especially those who hold leadership positions in industry, can drive new opportunities for collaboration. Additionally, participation in trade missions offers further avenues to grow UI relationships by fostering direct engagement with industry leaders on shared interests.
5. **Intentional Relationship-Building/Strategic Approach:** Effective UI partnerships do not happen by chance—they require deliberate effort and active engagement. Success depends on sustained commitment to building strong relationships. A deliberate roadmap with short-, medium-, and long-term objectives is beneficial, recognizing differing goals between universities and companies.
6. **Regular Reviews:** Establishing a roadmap is not enough—ongoing check-ins and flexibility are crucial to reassess progress and adjust strategies to maintain alignment.
7. **Trust and Mutual Goals:** Consistent communication is fundamental. Partnerships thrive on trust, shared goals, and long-term planning, often starting small and expanding over time.

Resources²

UIDP Resources

- 15 effective practices for strong UI partnerships, illustrated with UIDP member case studies.* uidp.org/publication/15-effective-practices-for-strong-ui-partnerships/
- Catalyzing UI Engagement Workshop. (2013). *Meeting summary from workshop about catalyzing UI engagement to promote the nation's competitiveness.* uidp.org/publication/catalyzing-ui-engagement/
- Communicating university research strengths to industry partners quick guide, guidance on how university faculty and staff can proactively present research capabilities to industry partners.* uidp.org/publication/communicating-university-research-strengths/
- Networking with a purpose: The basics of facilitated serendipity, how to encourage serendipity in UI collaboration and the role 'connectors' play in identifying collaborative opportunities.* uidp.org/publication/networking-with-a-purpose/
- Partnership Continuum, a new way of thinking about the myriad ways in which academia and industry can interact for mutual benefit, often in ways which contribute to national growth.* uidp.org/publication/partnership-continuum/

UIIN Resources

- Strategic partnerships with industry,* University Industry Innovation Magazine. March 2021. <https://uiin.org/wp-content/uploads/2021/03/UIIM-Strategic-Partnerships-March21.pdf>

NACRO Resources

- Position paper: Elevating corporate relations through institutional commitment,* NACRO Benchmarking Committee, July 2016. nacrocon.org/publications
- White paper: Engagement of academic corporate relations officers in university-industry centers of research excellence,* NACRO Benchmarking Committee, July 15, 2015. nacrocon.org/publications
- White paper: Five essential elements of a successful twenty-first century university corporate relations program,* NACRO Benchmarking Committee, August 2, 2011. nacrocon.org/publications

²The work of connecting, exploring, uncovering, and building relationships is integral to building UI partnerships and fundamentally built on best-practices. There is value in reading about the collected experiences of professionals who have dedicated their careers to building and/or analyzing UI partnerships in support of research and innovation. The authors of the referenced material below are members of professional organizations, academic researchers, technology licensing, and corporate engagement professionals. Despite the range of experiences and viewpoints of the authors, the references provide convergent perspectives on UI partnerships, with an overwhelming agreement that success requires facilitators (not just tools), aligned interests that are multi-pronged.

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